

In-Class Quiz: May 14

Points: 8/8

✓ **Correct** 1/1 Points

1. In a **fully associative cache**, a memory block can be placed: 

- ☐ Only in one fixed location
- ☒ In any cache entry
- ☐ Only in L1 cache
- ☐ Only in main memory

✓ **Correct** 1/1 Points

2. In a **direct mapped cache**, a memory block maps to: 

- ☐ Any location
- ☐ Multiple locations
- ☒ Exactly one location
- ☐ Only L2 cache

✓ **Correct** 1/1 Points

3. The index in a direct mapped cache is typically computed using: 

- ☐ Tag bits

- ☐ Offset bits
- ☒ Block address modulo number of entries
- ☐ Random selection

✓ **Correct** 1/1 Points

4. A **set associative cache** with k blocks per set is called: 

- ☐ k-level cache
- ☐ k-bit cache
- ☒ k-way associative cache
- ☐ k-index cache

✓ **Correct** 1/1 Points

5. In a k-way set associative cache, how many locations are checked during lookup? 

- ☐ 1
- ☒ k
- ☐ All cache entries
- ☐ None

✓ **Correct** 1/1 Points

6. In a fully associative cache, which component is typically used for tag comparison? 

- ☐ SRAM array
- ☐ DRAM cells

☒ CAM (Content Addressable Memory)

☐ Register file

✓ **Correct** 1/1 Points

7. In a typical cache hierarchy, what type of memory cells are generally used to build the L1 and L2 caches? 

☐ DRAM cells

☒ SRAM cells

☐ Master Slave D flip flops

☐ A mix of DRAM and flip flops

✓ **Correct** 1/1 Points

8. What happens when a cache miss occurs? 

☐ Data is discarded

☐ CPU halts permanently

☒ Data is fetched from lower memory levels and cache is updated

☐ Cache is cleared



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